AMANAI LAB · INTERVIEW PREPARATION SERIES

AI System Design
Complete Roadmap 2026

Foundations · HLD · LLD · ML System Design · GenAI · Agentic AI · MCP · A2A

42 Episodes | 5 Modules | 3 Design Types | 23 Practice Problems

The Journey — 5 Modules, 42 Episodes

Every module has practice problems woven in — no long concept-only blocks.

01	System Design Foundations	Ep 01–12	8C 4P
02	High Level Design — AI-Relevant	Ep 13–17	0C 5P
03	Low Level Design — OOP + Problems	Ep 18–27	4C 6P
04	ML, GenAI & Agentic AI	Ep 28–40	5C 8P
05	Grand Finale	Ep 41–42	0C 2P

EPISODE TYPES Concept | Practice

01 System Design Foundations

Core building blocks — concepts taught in pairs, followed immediately by a practice problem.

8C 4P

Ep 01	Concept	What is System Design · Interview Framework · Estimation & Requirements
Ep 02	Concept	Networking · DNS · CDN · Forward & Reverse Proxy · API Gateway
Ep 03	Concept	Load Balancer · Caching (Redis, CDN, Semantic) · Cache Invalidation
Ep 04	Concept	Databases — SQL vs NoSQL vs Vector DB · Blob Storage · When to use what for AI
Ep 05	Practice	Practice — Design URL Shortener : hashing, DB choice, read-heavy scaling
Ep 06	Concept	Message Queues (Kafka) · REST vs gRPC vs GraphQL · WebSockets vs SSE
Ep 07	Concept	Rate Limiting · Auth (OAuth 2.0, JWT, SSO) · CAP Theorem · PACELC Theorem
Ep 08	Practice	Practice — Design a Rate Limiter : token bucket, leaky bucket, sliding window
Ep 09	Concept	Sharding · Replication · Consistent Hashing · Microservices · Event-Driven Arch
Ep 10	Concept	Circuit Breaker · Fault Tolerance · Idempotency · Observability · Cost Reasoning
Ep 11	Practice	Practice — Design WhatsApp : real-time messaging, WebSockets, E2E encryption
Ep 12	Practice	Practice — Design Notification System : push, email, SMS, fan-out at scale

02 High Level Design (HLD) — AI-Relevant Problems

All practice. Five classic systems that directly map to AI engineering interviews.

0C 5P

Ep 13	Practice	Design YouTube — upload pipeline, transcoding, CDN, recommendation engine
Ep 14	Practice	Design Uber — geolocation, real-time driver matching, surge pricing
Ep 15	Practice	Design Twitter / X Feed — fanout strategies, timeline ranking, caching
Ep 16	Practice	Design Search Autocomplete — trie, prefix ranking, personalisation
Ep 17	Practice	Design DynamoDB / Key-Value Store — consistent hashing, quorum, replication

03 Low Level Design (LLD) — OOP + Patterns + Problems

4 concept episodes covering all patterns, then 6 real interview problems back to back.

4C 6P

Ep 18	Concept	OOP Pillars · SOLID Principles · UML Class Diagrams · Identifying classes
Ep 19	Concept	Design Patterns — Creational: Singleton, Factory, Builder, Prototype
Ep 20	Concept	Design Patterns — Behavioral: Strategy, Observer, State, Chain of Responsibility
Ep 21	Concept	Design Patterns — Structural: Decorator, Adapter, Proxy, Facade
Ep 22	Practice	LLD Practice — Parking Lot System : most-asked LLD across all companies
Ep 23	Practice	LLD Practice — BookMyShow : seat locking, concurrent booking, payment
Ep 24	Practice	LLD Practice — LRU / LFU Cache : LinkedList + HashMap, used in AI caching
Ep 25	Practice	LLD Practice — Splitwise : expense splitting, graph-based settlements
Ep 26	Practice	LLD Practice — Hotel Management System : rooms, booking, billing
Ep 27	Practice	LLD Practice — Ride Sharing App (Uber LLD) : matching, pricing, trips

04 ML, GenAI & Agentic AI System Design

The core of the series — ML concepts then practice, followed by GenAI concepts then practice.

7C 6P

Ep 28	Concept	ML SD Framework · Data Pipeline · Feature Store · How ML SD differs from HLD
Ep 29	Concept	Model Training Pipeline · Model Serving · Training-Serving Skew · Drift Monitoring
Ep 30	Practice	ML Practice — Design Netflix Recommendation System + Fraud Detection System
Ep 31	Practice	ML Practice — Design Google Search Ranking + Ad Click Prediction (Meta)
Ep 32	Concept	GenAI vs ML Systems · RAG Architecture · Chunking · Vector DB at Scale
Ep 33	Concept	Fine-tuning vs RAG vs Prompt Engineering · Hallucination Mitigation
Ep 34	Concept	LLM Inference Optimization · Semantic Caching · LLM Gateway · Token Budget
Ep 35	Concept	MCP Protocol (Agent→Tool) · A2A Protocol (Agent→Agent) · Multi-Agent Orchestration
Ep 36	Concept	AI Guardrails · Prompt Injection Defense · Evals (RAGAS, LLM-as-Judge) · Prompt Versioning
Ep 37	Practice	GenAI Practice — Design AI PDF Chatbot : full RAG system end to end
Ep 38	Practice	GenAI Practice — Design Multi-Agent Research System : orchestrator + sub-agents
Ep 39	Practice	GenAI Practice — Design AI Resume Analyzer + AI Interview Copilot (AmanAI Lab)
Ep 40	Practice	GenAI Practice — Design Perplexity-style AI Search + LLM Serving Infrastructure

05 Grand Finale

Close with AI-specific low level design, then a full live mock interview.

0C 2P

Ep 41	Practice	AI LLD — Agent Orchestrator class design + RAG Pipeline low-level design
Ep 42	Practice	Full Mock Interview — 45-minute live AI system design walkthrough